GrappStar launcher, desk size (0.3 of the design model), TEST BUILD: snap together, all printed. 14 part files, 21 printed pieces.
All sizes mm. Everything fits a 256 bed.

PARTS

01_gear.stl	59 x 63 x 17 mm	31 cm3	print: flat on the teeth, 100% infill
02_drive_gear.stl	42 x 42 x 17 mm	10 cm3	print: flat on the teeth, 100% infill
03_axle.stl	20 x 20 x 117 mm	16 cm3	print: standing on the head, 100% infill
04_carriage.stl	182 x 52 x 78 mm	45 cm3	print: standing on the rack end, supports on, 100% infill
05_dart.stl	20 x 20 x 69 mm	7 cm3	print: nose up
06_rail_body.stl	230 x 131 x 79 mm	209 cm3	print: on its side (barrel axis along the bed), supports on
07_frame.stl	156 x 118 x 159 mm	240 cm3	print: standing on the base plate
08_pinion.stl	43 x 43 x 14 mm	12 cm3	print: flat, 100% infill
09_crank_gear.stl	82 x 82 x 38 mm	65 cm3	print: flat on the gear face, stub up; 100% infill
14_crank_pin.stl	18 x 18 x 27 mm	3 cm3	print: standing on the head, 100% infill
10_conrod.stl	22 x 100 x 7 mm	9 cm3	print: flat, 100% infill
11_gear_pin.stl	18 x 18 x 49 mm	6 cm3	print: standing on the head, 100% infill
12_clip.stl	16 x 15 x 2 mm	0 cm3	print: flat, 100% infill
13_bushing.stl	18 x 18 x 16 mm	2 cm3	print: flat, 100% infill

TEST BUILD: everything printed, nothing bought except the motor. No one way bearings yet: three printed bushings sit where they go, the gears spin free on the axle.
Every pin is printed with a head on one end and a groove on the other; a printed C clip snaps into the groove. 6 clips, 3 bushings.

ASSEMBLY

1 Push a bushing into the drive gear, one into the pinion, one into the pedestal pocket from the inside face.

2 Rail body: big gear between the two bracket bosses, gear pin in from the minus side, clip on the end.

3 Crank gear: crank pin in from the back of the gear (head on the back), rod eye over it, clip. Stub into the pedestal bushing from the inside, clip on the outside. Other rod eye over the lug pin under the rail bracket, clip.

4 Axle: in from the plus side, through the hole in the motor plate (the head fits through it), then the plus leg, drive gear, both bracket pivot bosses, pinion, minus leg. The head stops against the leg, inside the motor plate. Clip on the minus end. Drive gear must mesh the big gear, pinion must mesh the crank gear.

5 No motor on this build: turn the axle head by hand. Forward cocks and fires, backward sets the pitch (once the one way bearings are in).

6 Carriage into the barrel from the breech, V ridges in the rail grooves, arm through the slit. Springs from the muzzle posts to the carriage pegs. Dart in at the muzzle.

CLIPS: the opening is 0.9 mm narrower than the groove core, push straight on, it snaps. Pull off with a flat blade. If one is too tight, scale it 102% in the slicer.
LATER: swap the three bushings for HF1216 one way bearings (same 18 x 12 x 16 size) and it becomes the working launcher.

PRINT: 0.2 mm layers. Gears, pins, clips, rod, bushings 100% infill. Rail body and frame 4 walls 30%. Supports where noted.